COMPUTER SCIENCE & IT

DIGITAL LOGIC



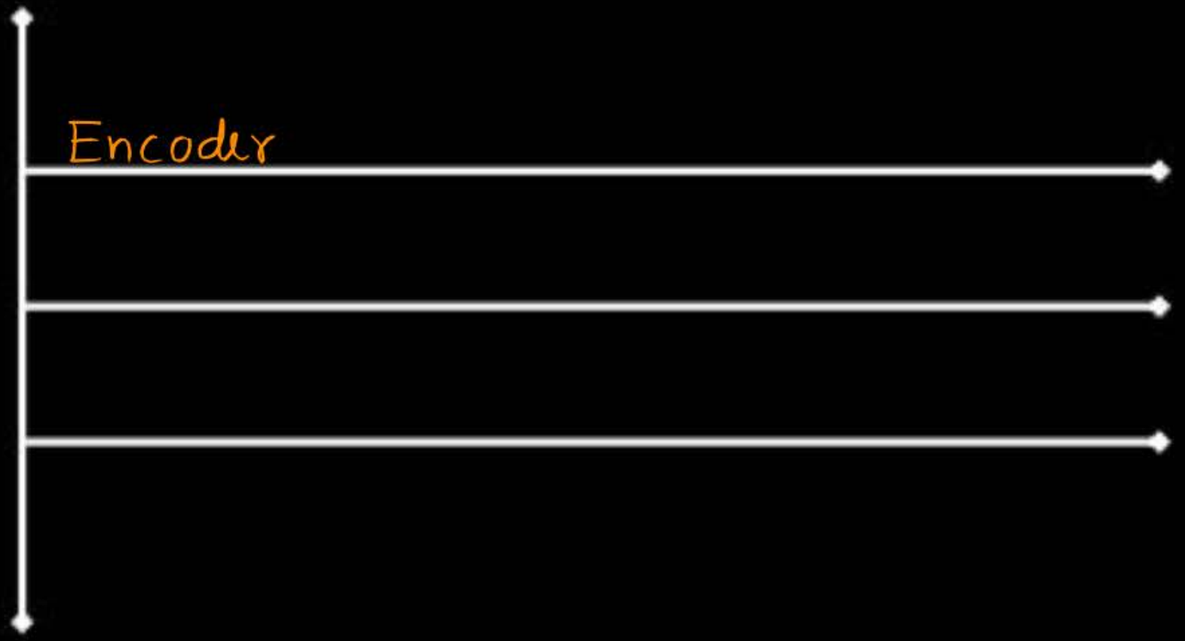
Lecture No. 14

Combinational Circuit



By- Chandan Gupta Sir


Recap of
Previous
Lecture


Encoder

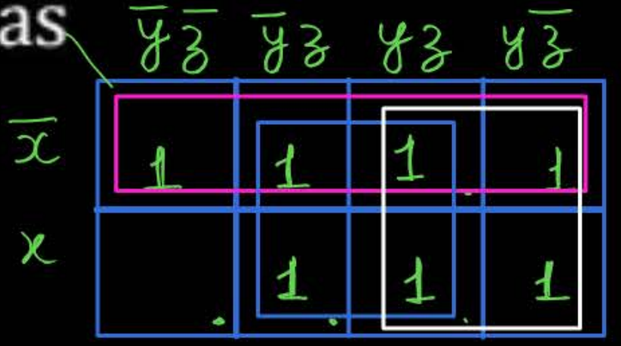


Question Discussion Cont.

Question

A logical function f(x, y, z) is given as

$$f(x,y,z) = x\bar{y} + yz$$



$= \pi(4)$

$= (\bar{x} + y + z)$

Another logical function

$f_1(x,y,z) = f[\bar{f}, f, z]$ will be

a. $\bar{x} + y + z$

b. $\bar{x}y + yz$

c. $(x + \bar{y})(\bar{y} + \bar{z})$

d. $(x + \bar{y})(\bar{y} + \bar{z})(y + z)$

$= \bar{f} \cdot \bar{f} + fz = \bar{f} + (f \cdot z) = (\bar{f} + f) \cdot (\bar{f} + z)$

$= \overline{x\bar{y} + yz} + z$

$= \bar{f} + z$

Question

A 3-input majority gate is defined by the logic function M (a, b, c) = ab + bc + ca. Which one of the following gate is $M\left(\overline{M(a,b,c)}, M(a,b,\bar{c}), c\right)$?

(a) 3-input NAND gate

(b) 3-input XOR gate

(c) 3-input NOR gate

(d) 3-input XNOR gate

$M\left[\overline{M(a,b,c)}, M(a,b,\bar{c}), c\right] = f(a,b,c)$

$f(0,0,0) = M[1, 0, 0] = 0$

$f(0,0,1) = M[1, 0, 1] = 1$

$f(0,1,0) = M[1, 1, 0] = 1$

$f(0,1,1) = M[0, 0, 1] = 0$

$M[1,0,0] = M[1,1,0] = 1$

$M[1,0,1] = M[0,0,1] = 0$

$M[1,1,0] = M[0,1,0] = 0$

$M[1,1,1] = M[0,1,1] = 1$

$f(a,b,c) = \Sigma(1,2,4,7) = a \oplus b \oplus c$

[NAT]

Consider the Karnaugh map given below, where X represents "don't care" and blank represents 0.

dc \ ba	00	01	11	10
00		x	x	
01	1			x
11	1			1
10		x	x	

H·W

Assume for all inputs(a, b, c, d) , the respective complements $(\bar{a}, \bar{b}, \bar{c}, \bar{d})$ are also available. The above logic is implemented using 2-input NOR gates only. The minimum number of gates required is _____.

[GATE-2017-CS: 1M]

[MCQ]

Consider the following Boolean expression for

F: F(P, Q, R, S) = PQ + P'QR + P'QR'S

The minimal sum-of-products form of F is

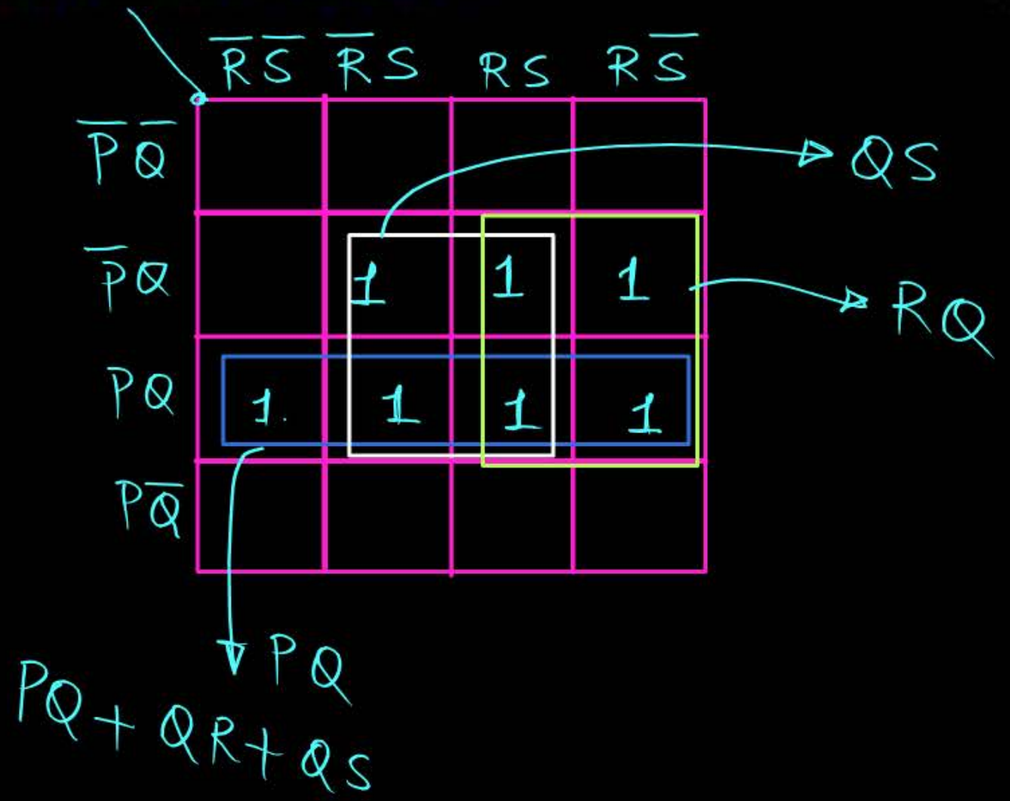
[GATE-2014-CS: 1M]

A. $PQ + QR + QS$

B. $P + Q + R + S$

C. $\overline{P} + \overline{Q} + \overline{R} + \overline{S}$

D. $\overline{P}R + \overline{P}\overline{R}S + P$



[MCQ]

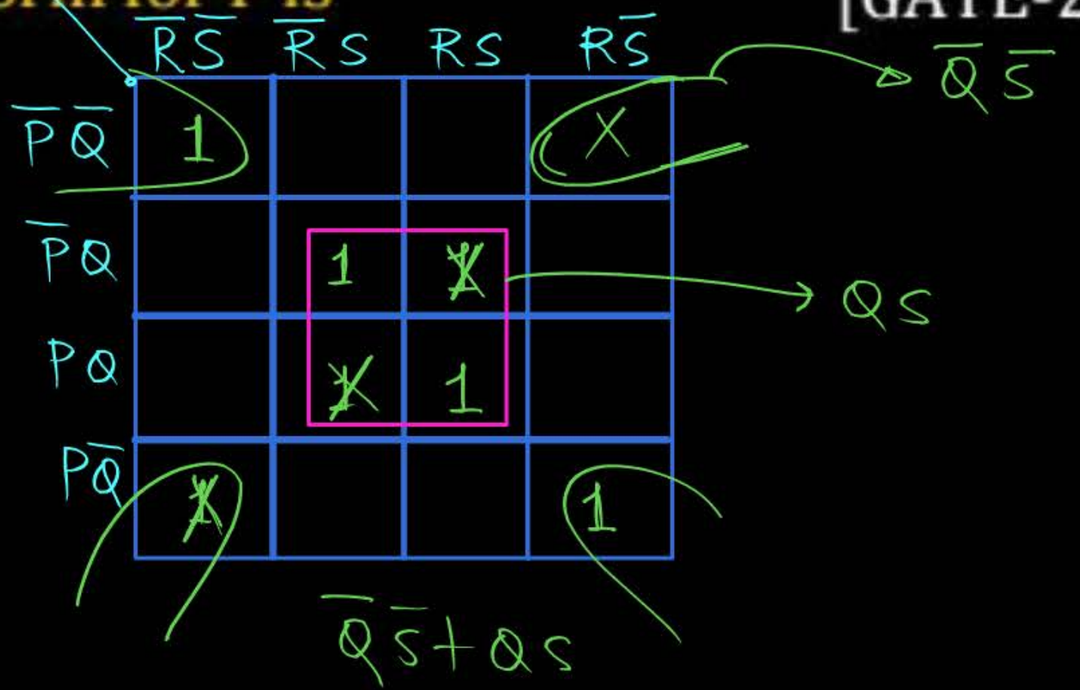
Consider the following minterm expression for

F: F (P,Q,R,S) = $\sum$ 0, 2, 5, 7, 8,10,13,15

The minterms 2, 7, 8 and 13 are 'do not care' terms.

The minimal sum-of-products form for F is

[GATE-2014-CS: 1M]



A $Q\bar{S} + \bar{Q}S$

B $\bar{Q}\bar{S} + QS$

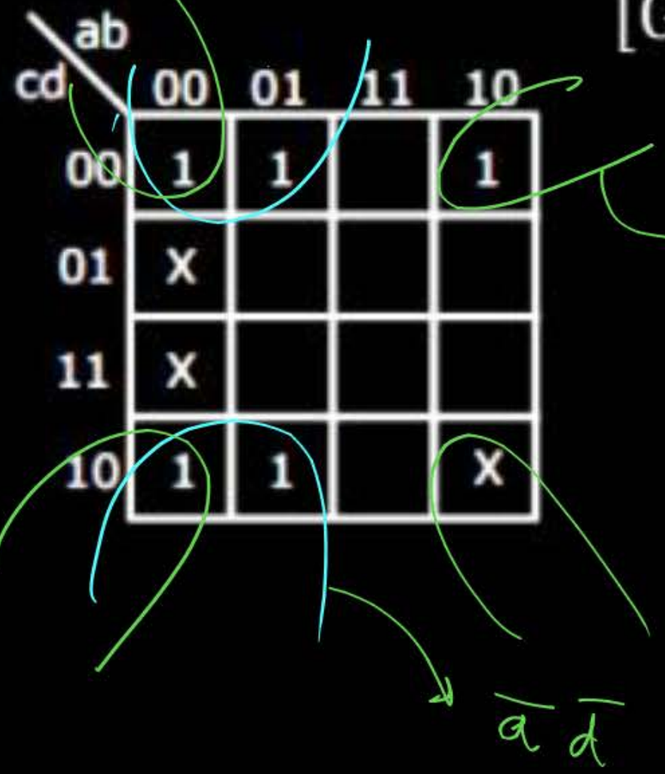
C $\bar{Q}R\bar{S} + \bar{Q}RS + Q\bar{R}\bar{S} + QRS$

D $\bar{P}Q\bar{S} + \bar{P}QS + PQS + P\bar{Q}\bar{S}$

[MCQ]

In the Karnaugh map shown below, X denotes a don't care term. What is the minimal form of the function represented by the Karnaugh map?

[GATE-2008-CS: M]



A $\bar{b}.\bar{d} + \bar{a}.\bar{d}$

B $\bar{a}.\bar{b} + \bar{b}.\bar{d} + \bar{a}.b.\bar{d}$

C $\bar{b}.\bar{d} + \bar{a}.b.\bar{d}$

D $\bar{a}.\bar{b} + \bar{b}.\bar{d} + \bar{a}.\bar{d}$

[MCQ]

Consider the following Boolean function of four variables f(A, B, C, D) = Σ(2, 3, 6, 7, 8, 9, 10, 11, 12, 13) The function is

[GATE-2008-CS: 1M]

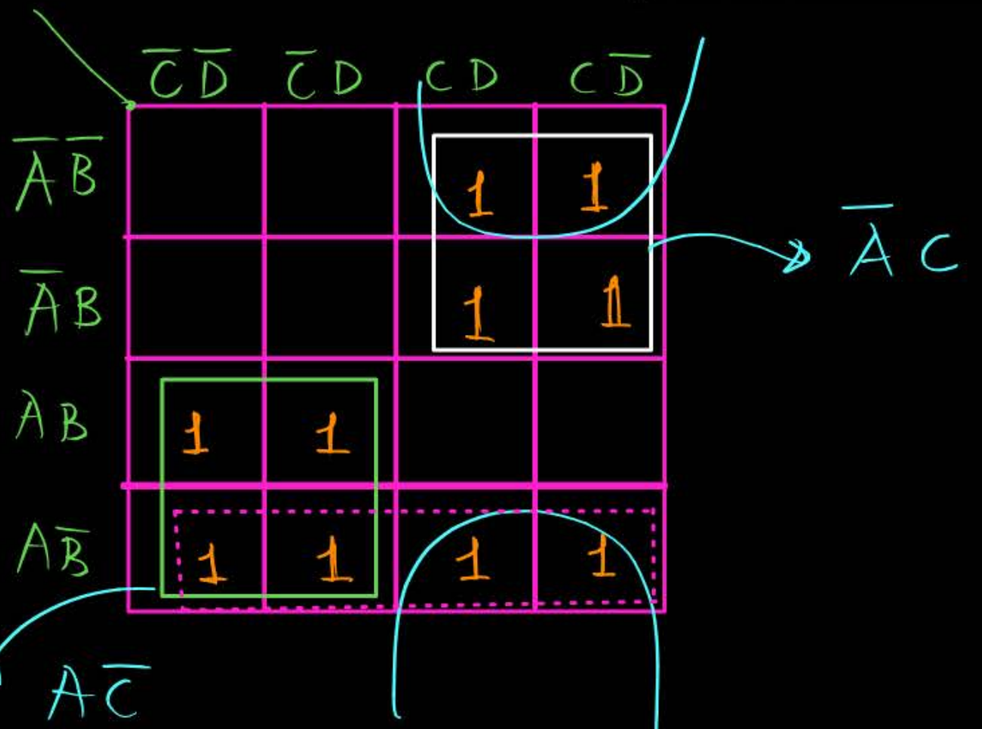
(A) independent of one variable

(B) Independent of two variables

(C) Independent of three variables

(D) Dependent on all the variables

No. of literal in minimized expression of f 5.



$= A\bar{C} + \bar{A}C + \bar{B}C$

or

$A\bar{C} + \bar{A}C + A\bar{B}$

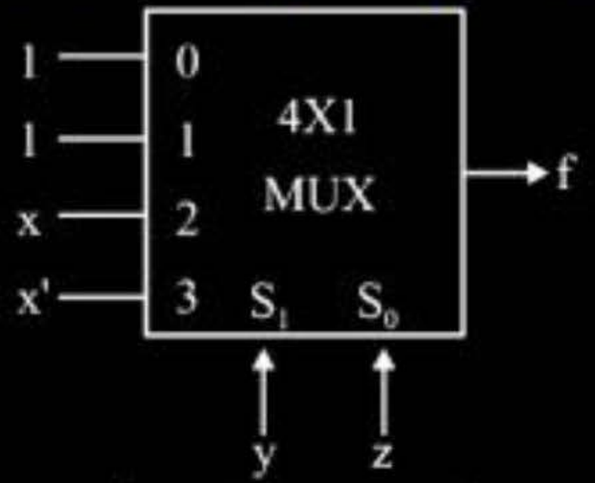
[MCQ]

Which one of the following circuits implements the Boolean function given below ?

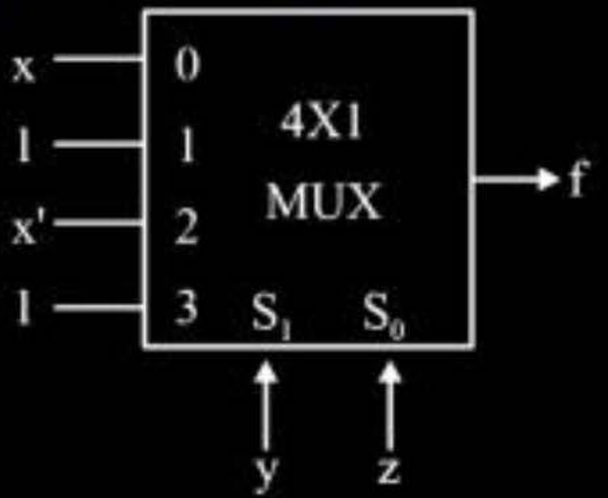
$f(x, y, z,) = m_0 + m_1 + m_3 + m_4 + m_5 + m_6$ where m_t is the ith minterm.

[GATE-2021-CS: 1M]

A

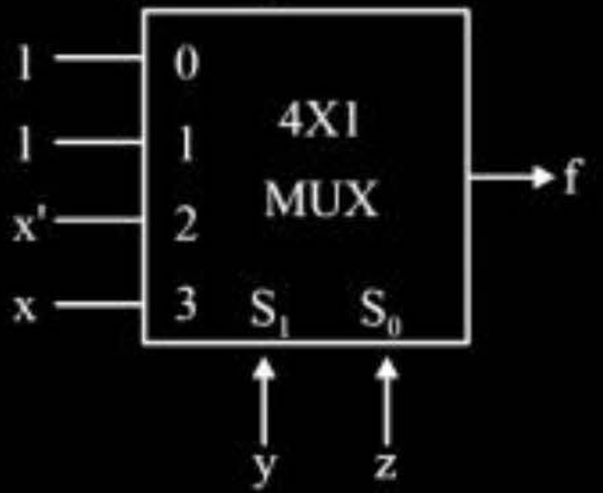


B

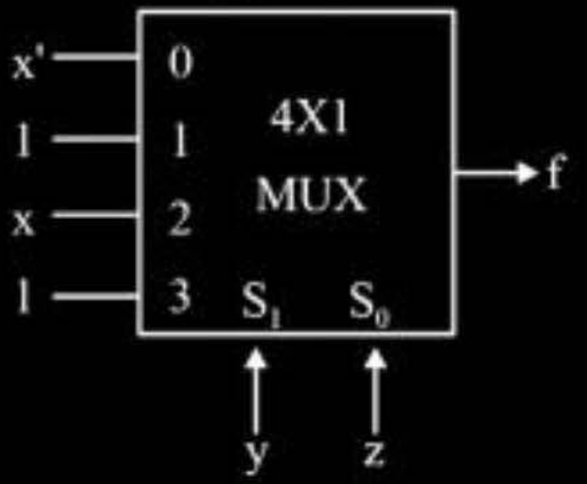


H·W

C



D



Question

A digital circuit is as shown below:

When $f_1(A, B) = \Sigma(0, 1)$ $f_2(A, B) = \Sigma 1 = f_3(A, B) = \Sigma(0,2,3)$,

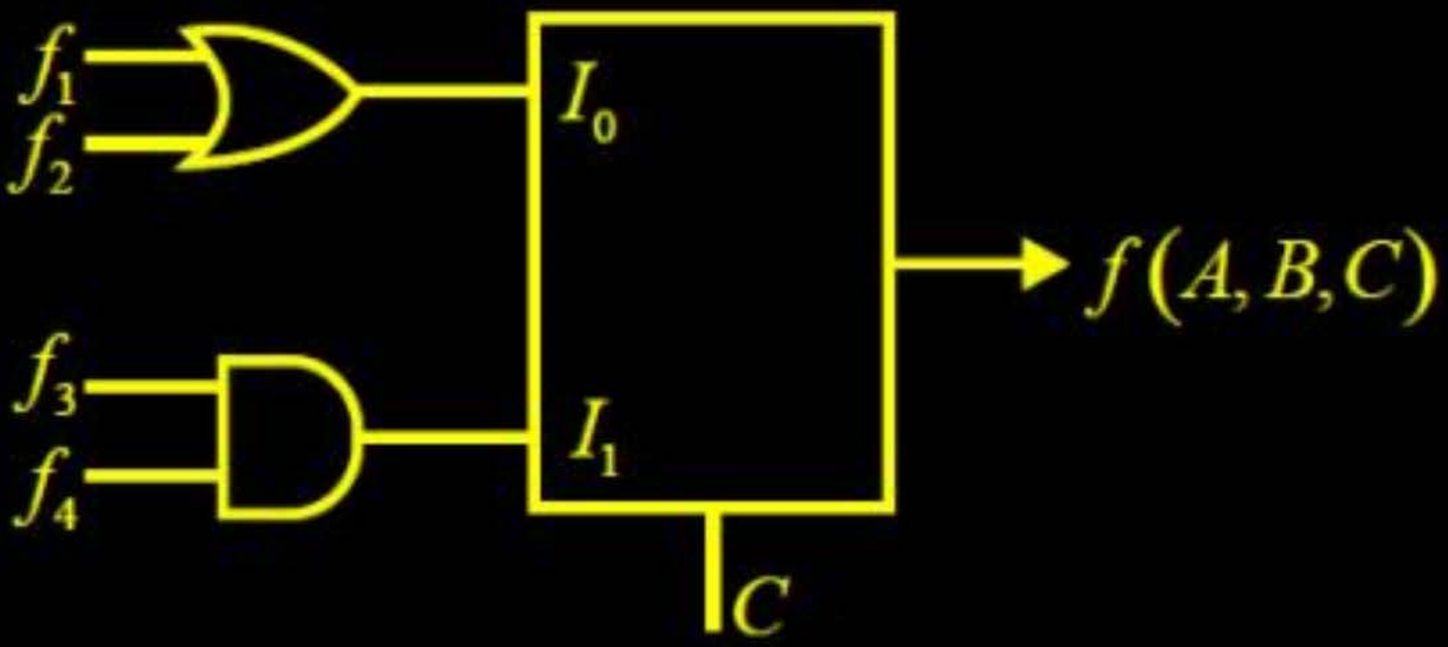
$f_4(A, B) = \Sigma(1,2,3)$ then $f(A, B, C)$ will be

(a) $A \oplus C$

(b) $A \odot C$

(c) $A \oplus B \oplus C$

(d) $A \odot B \oplus C$



[NAT]

A multiplexer is placed between a group of 32 registers and an accumulator to regulate data movement such that at any given point in time the content of only one register will move to the accumulator. The number of select lines needed for the multiplexer is _____.

[GATE-2020-CS: 1M]

H.W

Question

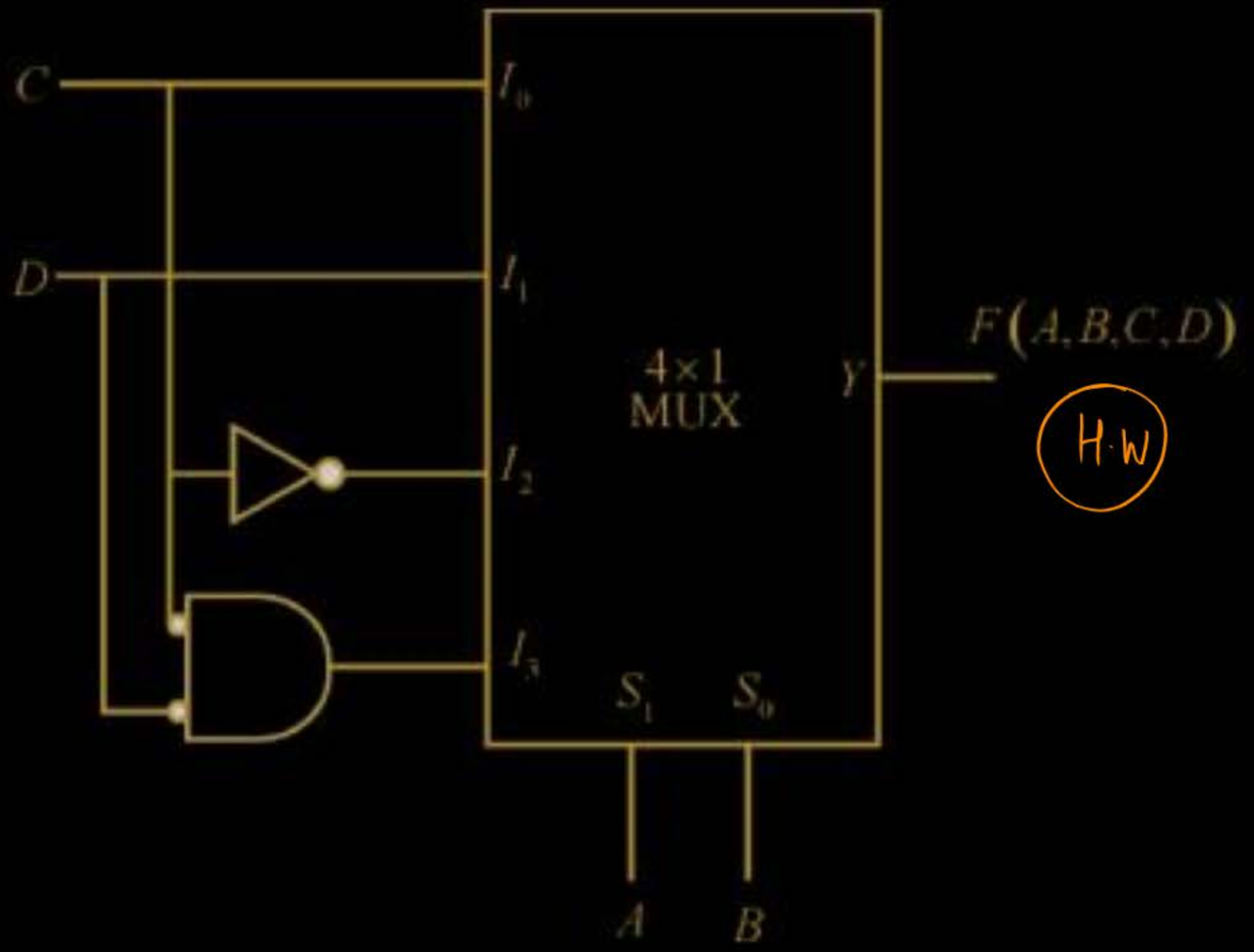
The Boolean function realized by the logic circuit shown is

(a) F = Σm(0,l,3,5,9,10,14)

(b) F = Σm(2,3,5,7 ,8,12,13)

(c) F = Σm(l,2,4,5,ll,14,15)

(d) F = Σm(2,3,5,7,8,9,12)



[MCQ]

Consider the 4 – to – 1 multiplexer with two select S_1 and S_0 given below.

The minimal sum of products from the Boolean expression for the output F of the multiplexer is -

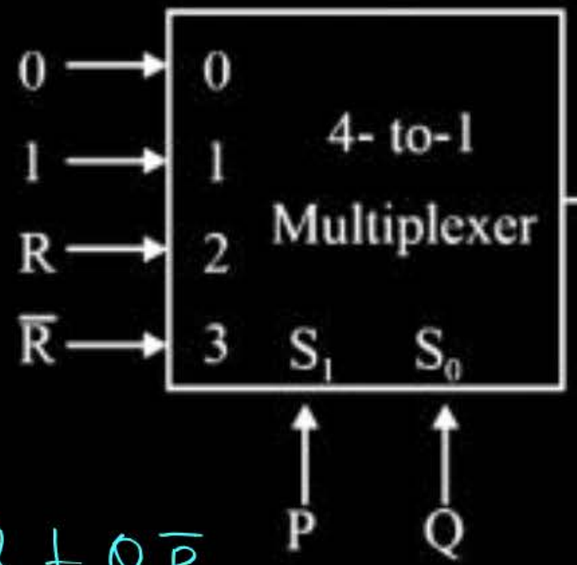
[GATE-2014-CS: 2M]

A. $\bar{P}Q + Q\bar{R} + P\bar{Q}R$

B. $\bar{P}Q + \bar{P}Q\bar{R} + PQ\bar{R} + P\bar{Q}R$

C. $PQR + \bar{P}Q\bar{R} + Q\bar{R} + P\bar{Q}R$

D. $PQ\bar{R}$



Question

Logical circuit is implemented as

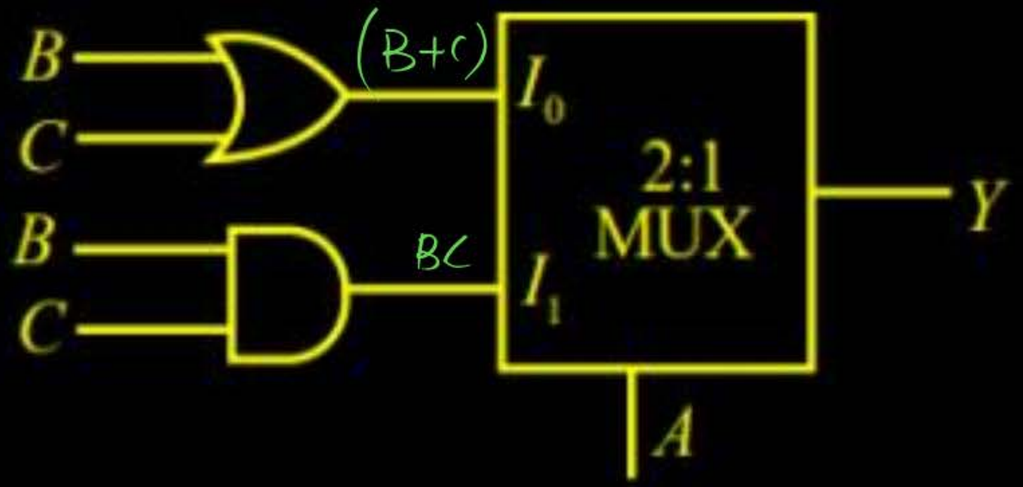
Y represents

(A) Borrow output of full subtractor

(B) Carry output of full adder

(C) 3-input XOR gate

(D) 3-input NOR gate



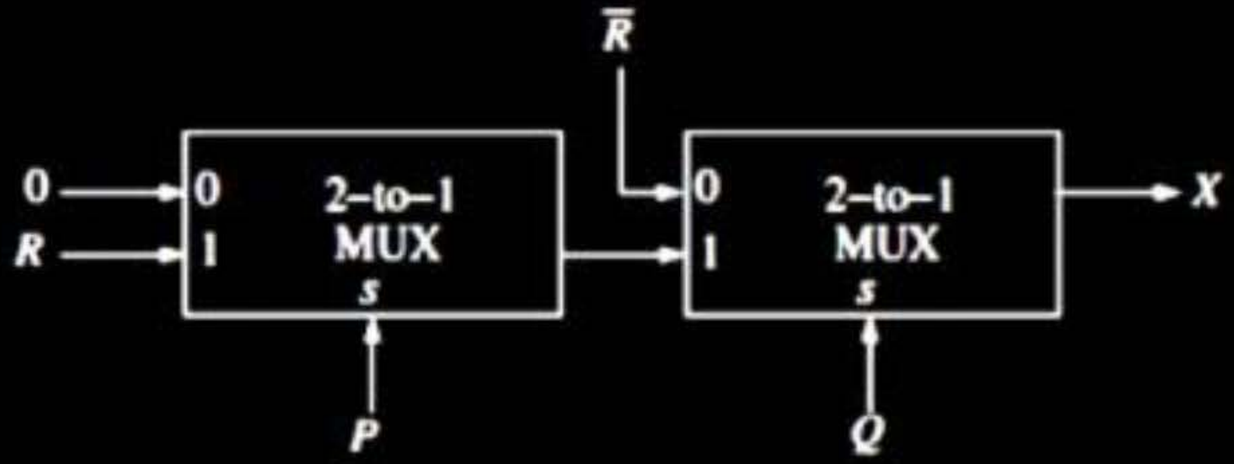
$= \bar{A}(B+C) + ABC$

$= \bar{A}B + \bar{A}C + ABC = \bar{A}B + \bar{A}C + BC$

$C[\bar{A} + AB] = C[\bar{A} + B]$

[MCQ]

Consider the two cascaded 2 to 1 multiplexers as shown in the figure.
The minimal sum of products form of the output X is : [GATE-2016-CS: 1M]



H.W

(A) $\overline{P}\overline{Q} + PQR$

(B) $\overline{PQ} + QR$

(C) $PQ + \overline{P}\overline{QR}$

(D) $\overline{QR} + PQR$

Question

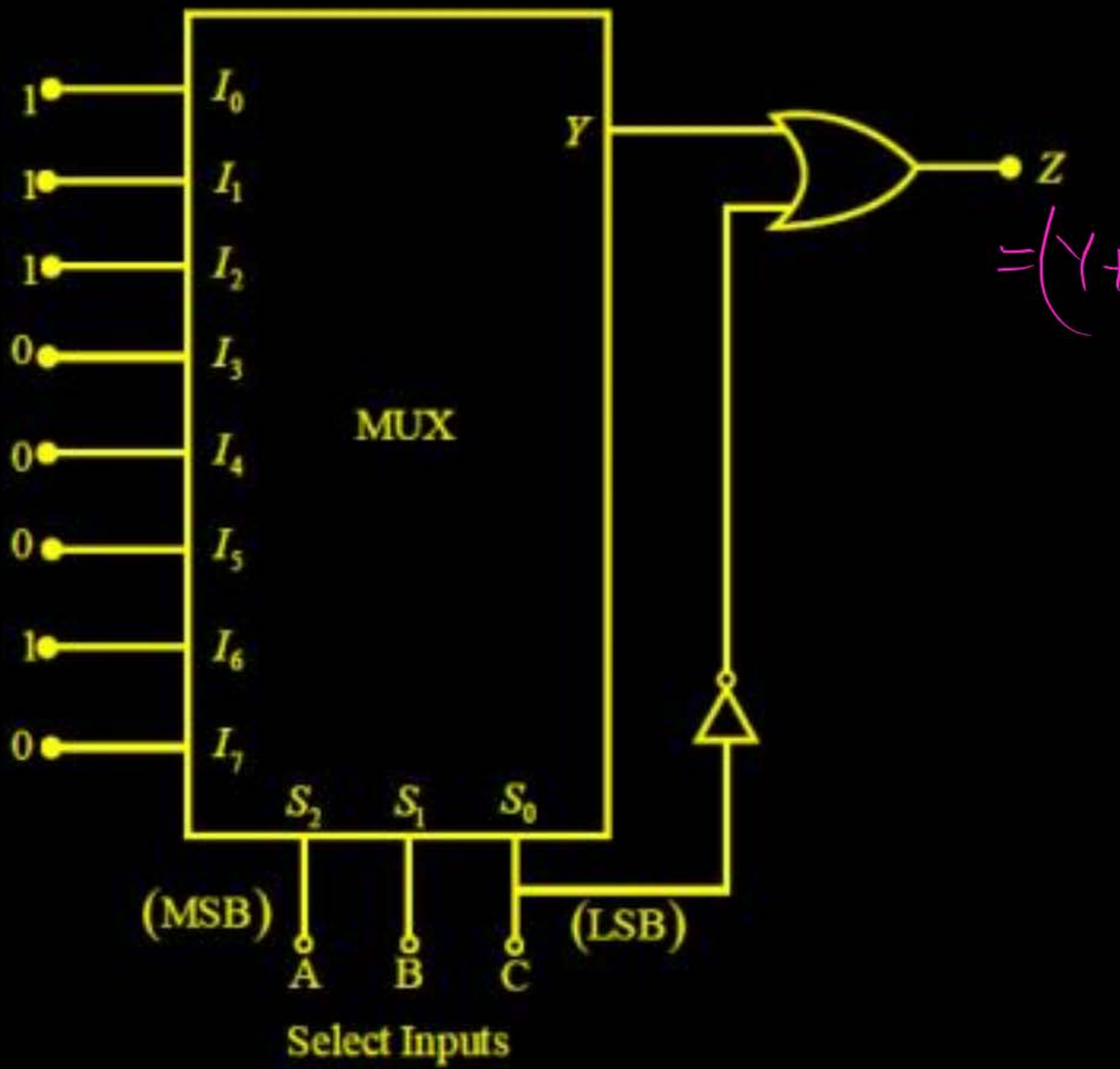
A combinational circuit using an 8 to 1 multiplexer is shown in the following figure. The minimized expression for the output (Z) is

(a) $C(\overline{A}+\overline{B})$

(b) $C(A+B)$

(c) $\overline{C}+\overline{A}\overline{B}$

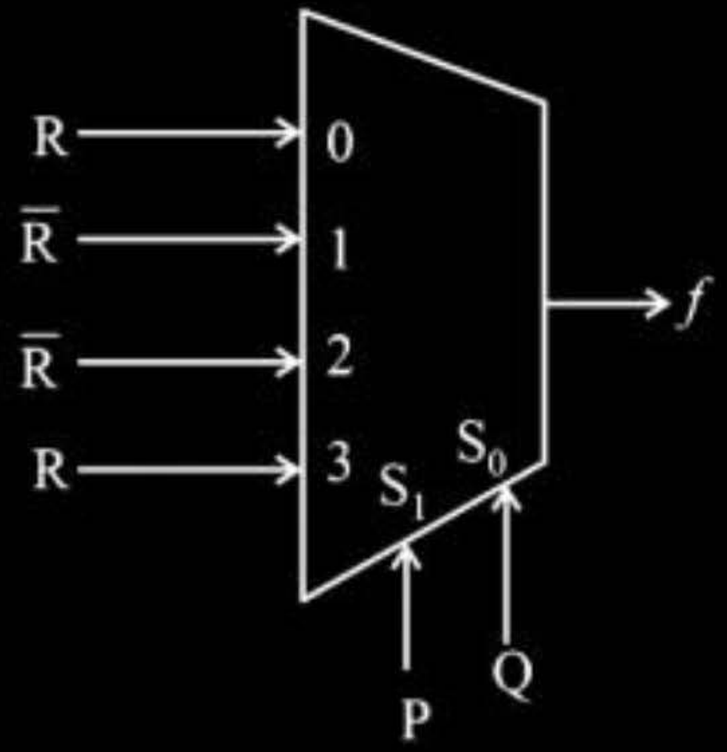
(d) $\overline{C}+AB$



[MCQ]

The Boolean expression of the output f of the multiplexer shown below is :

[GATE-2010-CS: 1M]



A $\overline{P \oplus Q \oplus R}$

B $P \oplus Q \oplus R$

C $P+Q+R$

D $\overline{P+Q+R}$

H.W

Question

logical function f (A, B, C) is given as

$$f(A,B,C)=\overline{AB+BC+CA}$$

Then $f_1(A,B,C)=f[AB,BC,CA]=\overline{ABC+\bar{A}BC+A\bar{B}C}=\overline{ABC}$

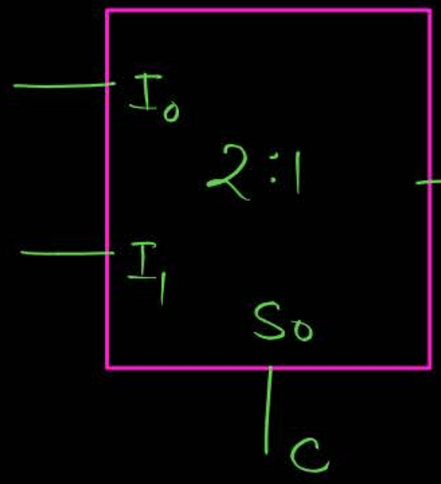
f_1 implemented using 2 : 1 MUX with C as select input then input I_0 & I_1 of 2: 1 MUX will be

(a) $I_0=1, I_1=\bar{A}$

(b) $I_0=0, I_1=\bar{A}+\bar{B}$

(c) $I_0=1, I_1=\bar{A}+\bar{B}$

(d) $I_0=0, I_1=\bar{A}+B$



$f_1(A,B,0)=1 \rightarrow I_0=1$

$f(A,B,1)=\overline{AB}=\bar{A}+\bar{B}$

$I_1=\bar{A}+\bar{B}$

Question

A bulb in a staircase has two switches, one switch being at the ground floor and the other one at the first floor. The bulb can be turned ON and also can be turned OFF by any one of the switches irrespective of the state of the other switch. The logic of switching of the bulb resembles

(a) an AND gate

(b) an OR gate

(c) an XOR gate

(d) a NAND gate

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

$Y = A \oplus B$

A	B	
0	0	1
0	1	0
1	0	0
1	1	1

$A \odot B$

Question

If $X = 1$ in the logic equation $[X + Z\{\bar{Y} + (\bar{Z} + X\bar{Y})\}]\{\bar{X} + \bar{Z}(X + Y)\} = 1$ then

(a) $Y = Z$

(b) $\bar{Y} = Z$

(c) $Z = 1$

(d) $Z = 0$

1 + anything

$1 \cdot$

$\bar{Y}Z = 1$

$\bar{Y} = 1$ & $Z = 1$

$1 \cdot (0 + \bar{Z}) = 1$

$\bar{Z} = 1$

$Z = 0$

$Y \odot Z = 1$

$Y = Z$

$Y \oplus Z = 1$

$Y = \bar{Z}$

$Y = 0$ or $Z = 1$ $\left(\bar{Y} + Z\right) = 1$

$Y = 0$, & $Z = 1$

or

$Y = 0$, & $Z = 0$

or

$Y = 1$, & $Z = 1$

Question

The minimal sum of products form of $f = A\bar{B}CD + \bar{A}BC + BCD + \bar{A}\bar{B}C$ is

(a) $\bar{A}C + BC$

(b) $\bar{A}C + CD$

(c) $AC + \bar{B}D$

(d) $A\bar{B} + C\bar{D}$

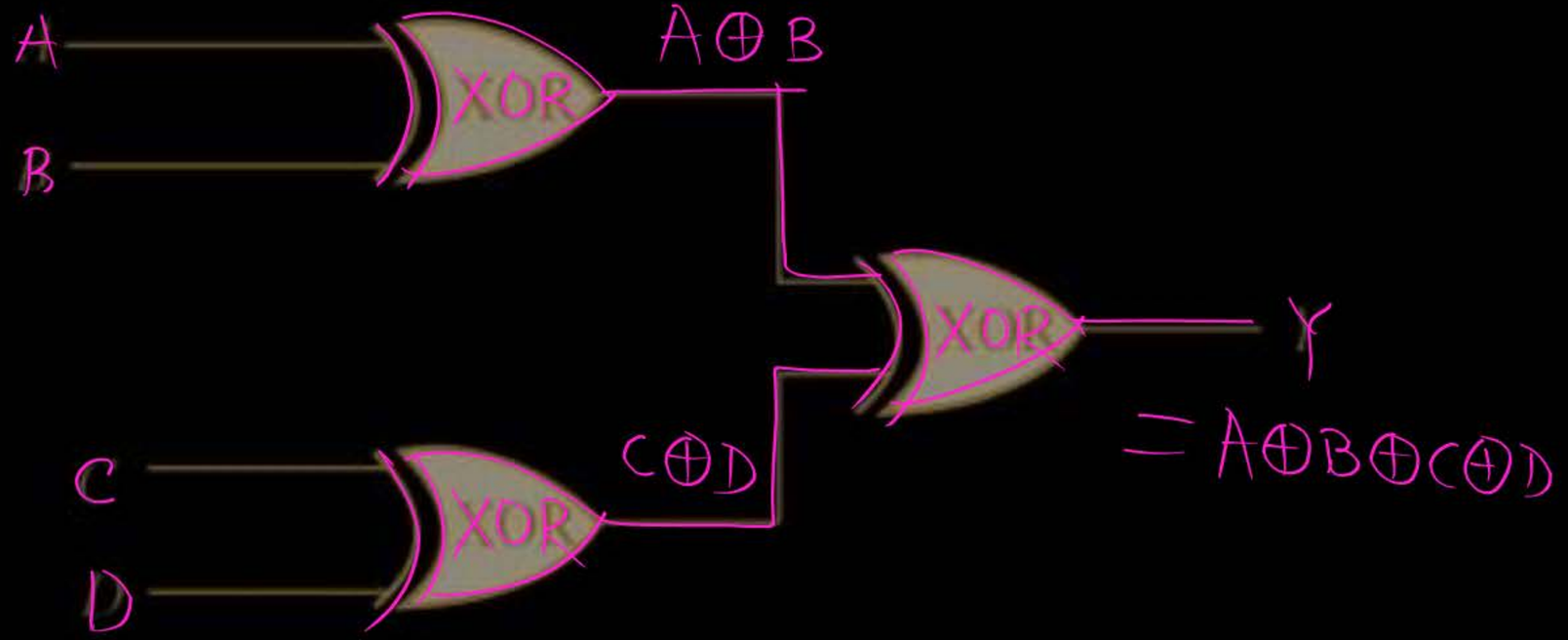
H.W

Question

A, B, C and D are input bits and Y is the output bit into the XOR gate circuit of the figure below. Which of the following statements about the sum S of A, B, C, D and Y is correct?

$S = (A + B + C + D + Y)$

addition



$S = A+B+C+D+Y$

(0, 1, 2, 3, 4)

(a) S is always either zero or odd.

(b) S is always either zero or even.

(c) S = 1 only if the sum of A, B, C and D is even.

(d) S = 1 only if the sum of A, B, C and D is odd.

$0 \longrightarrow Y = 0 \longrightarrow S = 0$

$1 \longrightarrow Y = 1 \longrightarrow S = 2$

$2 \longrightarrow Y = 0 \longrightarrow S = 2$

$3 \longrightarrow Y = 1 \longrightarrow S = 4$

$4 \longrightarrow Y = 0 \longrightarrow S = 4$

Question

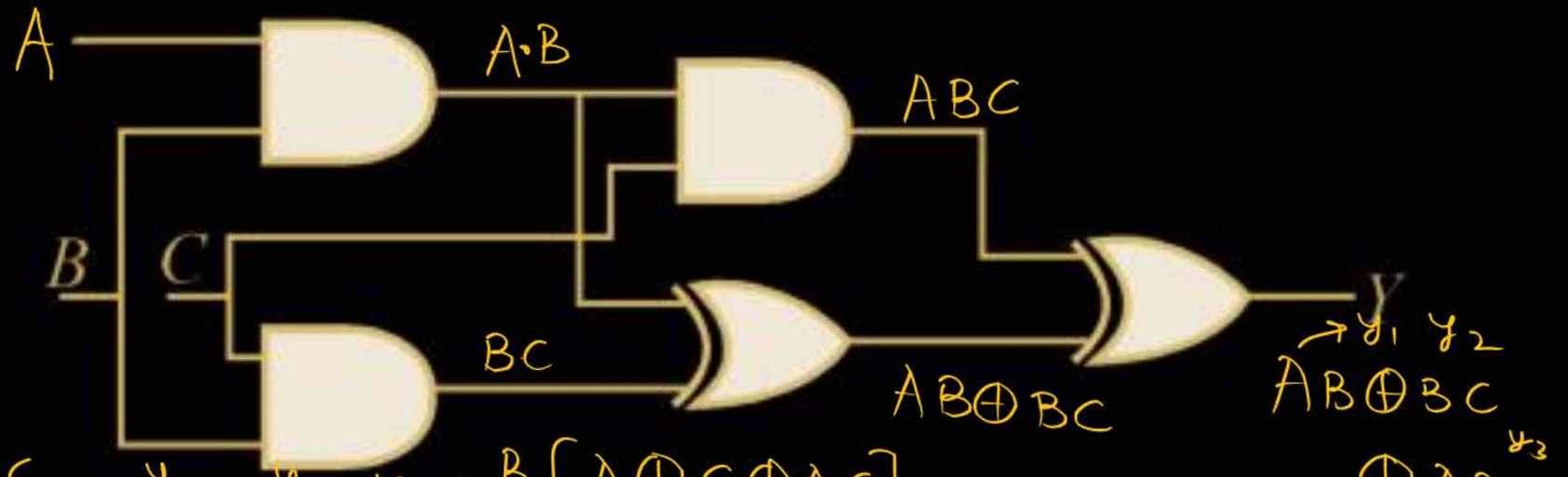
The input of the combinational circuit given below is

(a) A + B + C

(b) A(B + C)

(c) B(C + A)

(d) C(A + B)



A B C y_1 y_2 y_3 y $B[A \oplus C \oplus AC]$

$B[A+C]$

$= P \oplus Q \oplus P \cdot Q$

$= P + Q$

$= AB + BC$

$= B[A+C]$

Question

A logical function $f(x,y,z) = \overline{\bar{x}y + \bar{y}z + xyz} + \bar{x}y\bar{z}$

Simplified output f(x, y, z) will be

(a) $y + z$

(b) $\bar{z}$

(c) $x\bar{y} + yz$

(d) $(y + \bar{z})(x + \bar{y} + \bar{z})$

Question

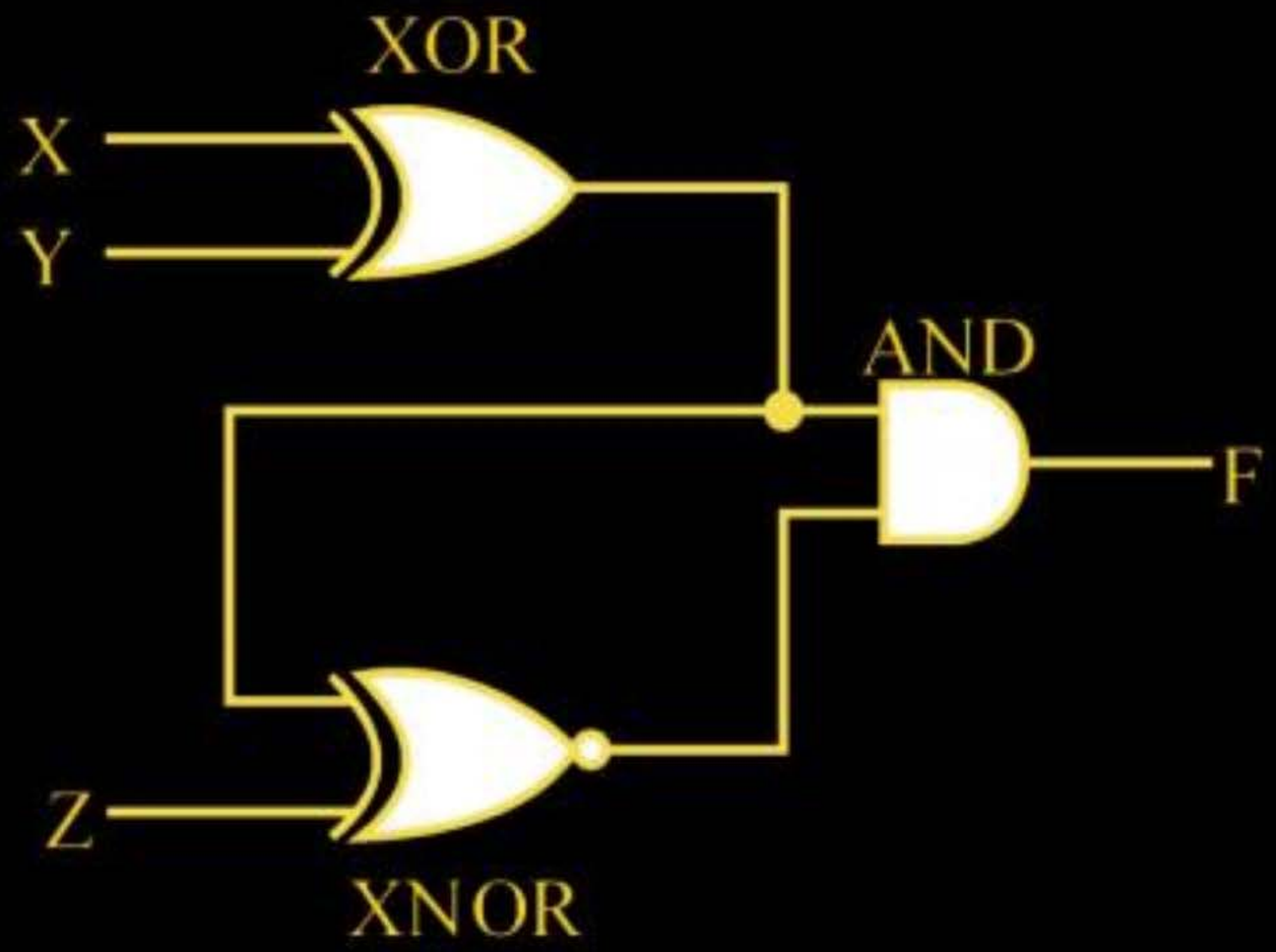
The output F in the digital logic circuit shown in the figure is

(a) $F = \bar{X}YZ + X\bar{Y}Z$

(b) $F = \bar{X}\bar{Y}Z + X\bar{Y}\bar{Z}$

(c) $F = \bar{X}\bar{Y}Z + XYZ$

(d) $F = \bar{X}Y\bar{Z} + XYZ$



H·W

Question

A logical expression $f(x,y,z)=\bar{x}y+xyz+x\bar{y}\bar{z}$

Then which of the following k-map represents the correct implementation of above logical function.

(a)

y \ z	$\bar{z}$	z
$\bar{y}$	x	
y	x	1

(b)

y \ z	$\bar{z}$	z
$\bar{y}$	x	
y	1	$\bar{x}$

(c)

y \ z	$\bar{z}$	z
$\bar{y}$	x	
y	$\bar{x}$	1

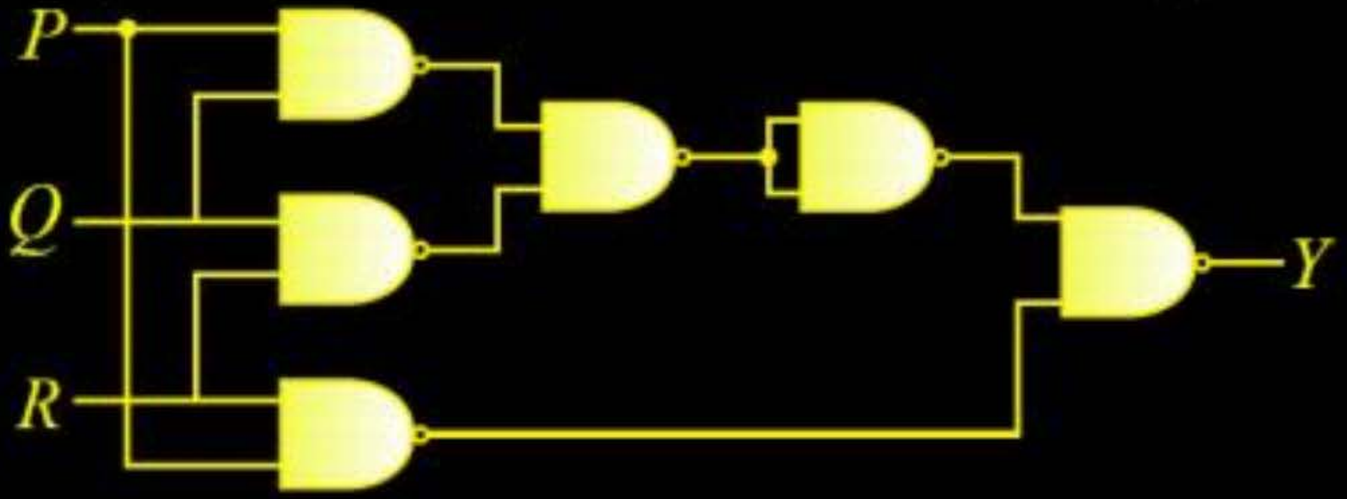
(d)

y \ z	$\bar{z}$	z
$\bar{y}$		x
y	$\bar{x}$	1

H.W

Question

The output Y is the circuit below is always "1" when



H.W

(a) Two or more of the input P, Q, R are "0".

(b) Two or more of the inputs P, Q, R are "1".

(c) Any odd number of the inputs P, Q, R are "0".

(d) Any odd number of the inputs P, Q, R are "1".

Question

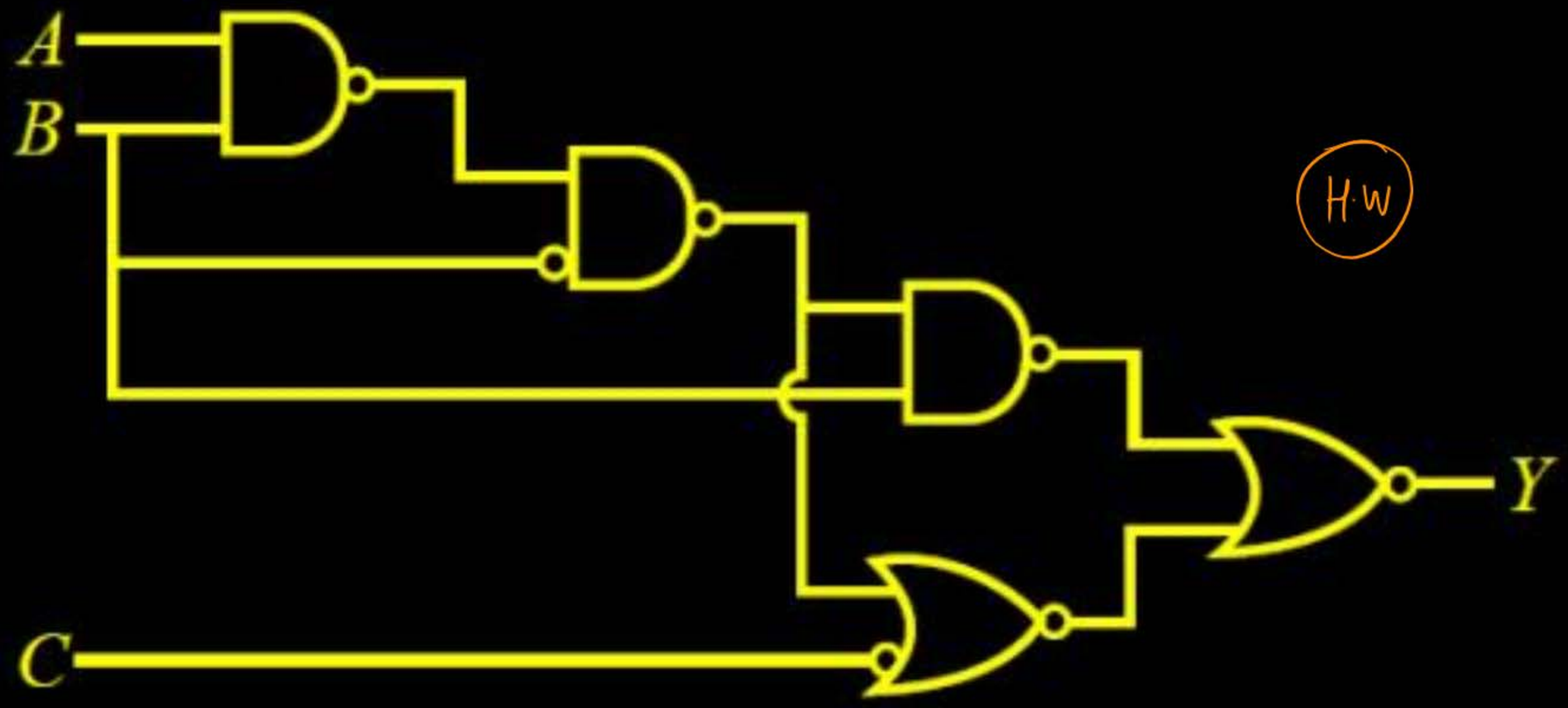
For the logic circuit shown in figure, the simplified Boolean expression for the output Y is

(a) A + B + C

(b) A

(c) B

(d) C



Question

K-map is given below:

	$\bar{B}\bar{C}$	$\bar{B}C$	BC	$B\bar{C}$
$\bar{A}$		1	X	
A	X	1	$\bar{D}$	

Where X-represents don't care condition.

The minimized solution of this K-map will be

(a) $C\bar{D}+\bar{B}C$

(b) $\bar{B}CD+B\bar{C}\bar{D}$

(c) $\bar{B}C+AC\bar{D}$

(d) $BC\bar{D}+A\bar{B}C$

Question

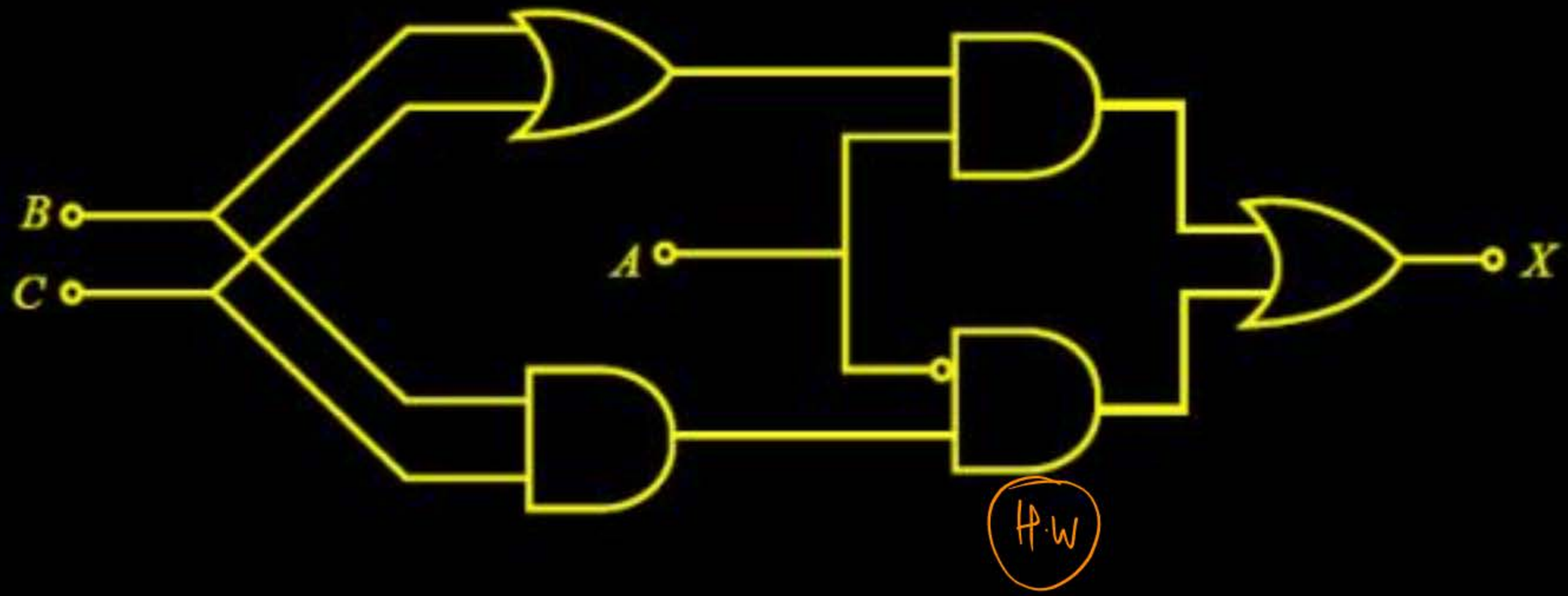
In the logic circuit shown is figure, the output X is

(a) $A\bar{B}+B\bar{C}+C\bar{A}$

(b) $A + B + C$

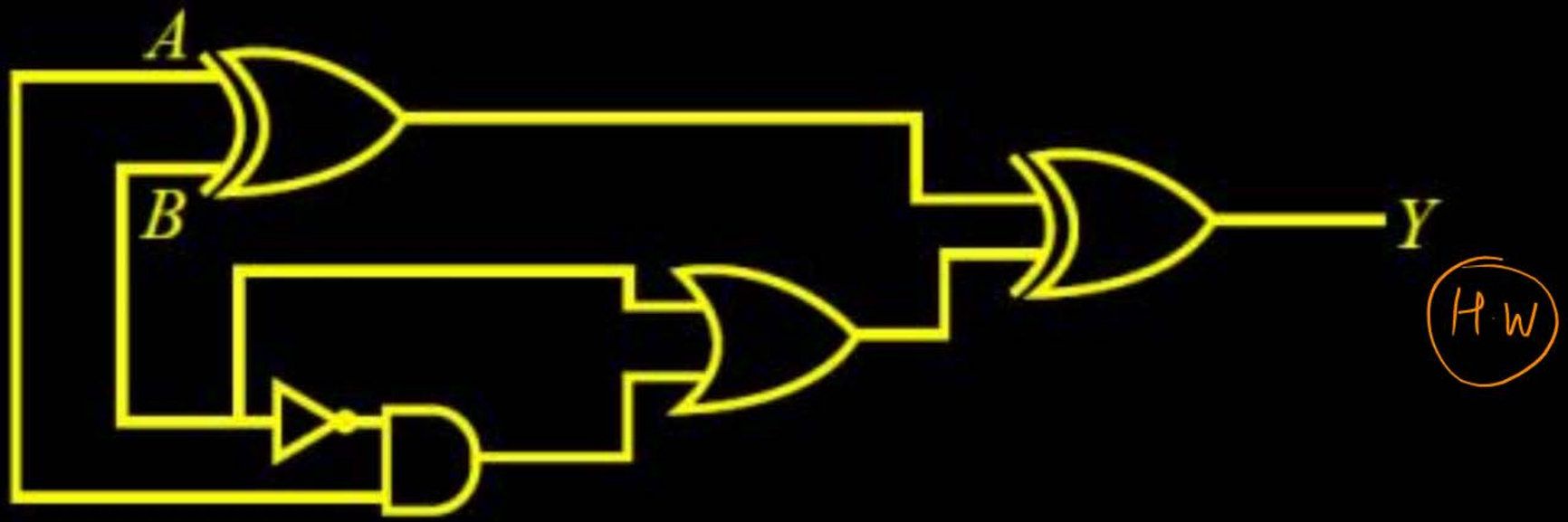
(c) $AB + BC + CA$

(d) $\overline{AB}+\overline{BC}+\overline{CA}$



Question

A logical circuit implemented as



To implement output Y minimum no. of 2:1 MUX required is

Question

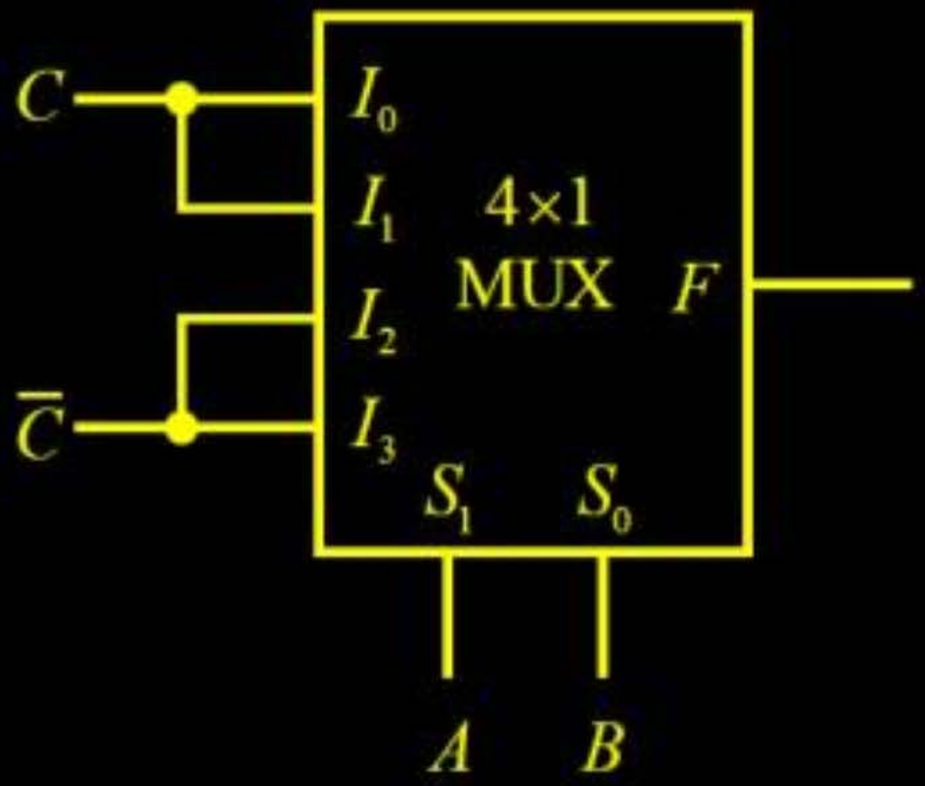
The logic realized by the circuited shown in figure is

(a) $F = A \odot C$

(b) $F = A \oplus C$

(c) $F = B \odot C$

(d) $F = B \oplus C$



H·W

Question

A K-map is given below

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	X	1	1	
$\bar{A}B$	1	X		
AB	1	X		
$A\bar{B}$		1	1	

Minimize form this K-map will be

(a) $B\bar{C}+\bar{B}D$

(b) $\bar{A}\bar{C}+\bar{C}D+\bar{B}D$

(c) $B\bar{C}+\bar{C}D+\bar{B}D$

(d) None of these

Question

Combination circuit in designed as shown below:-

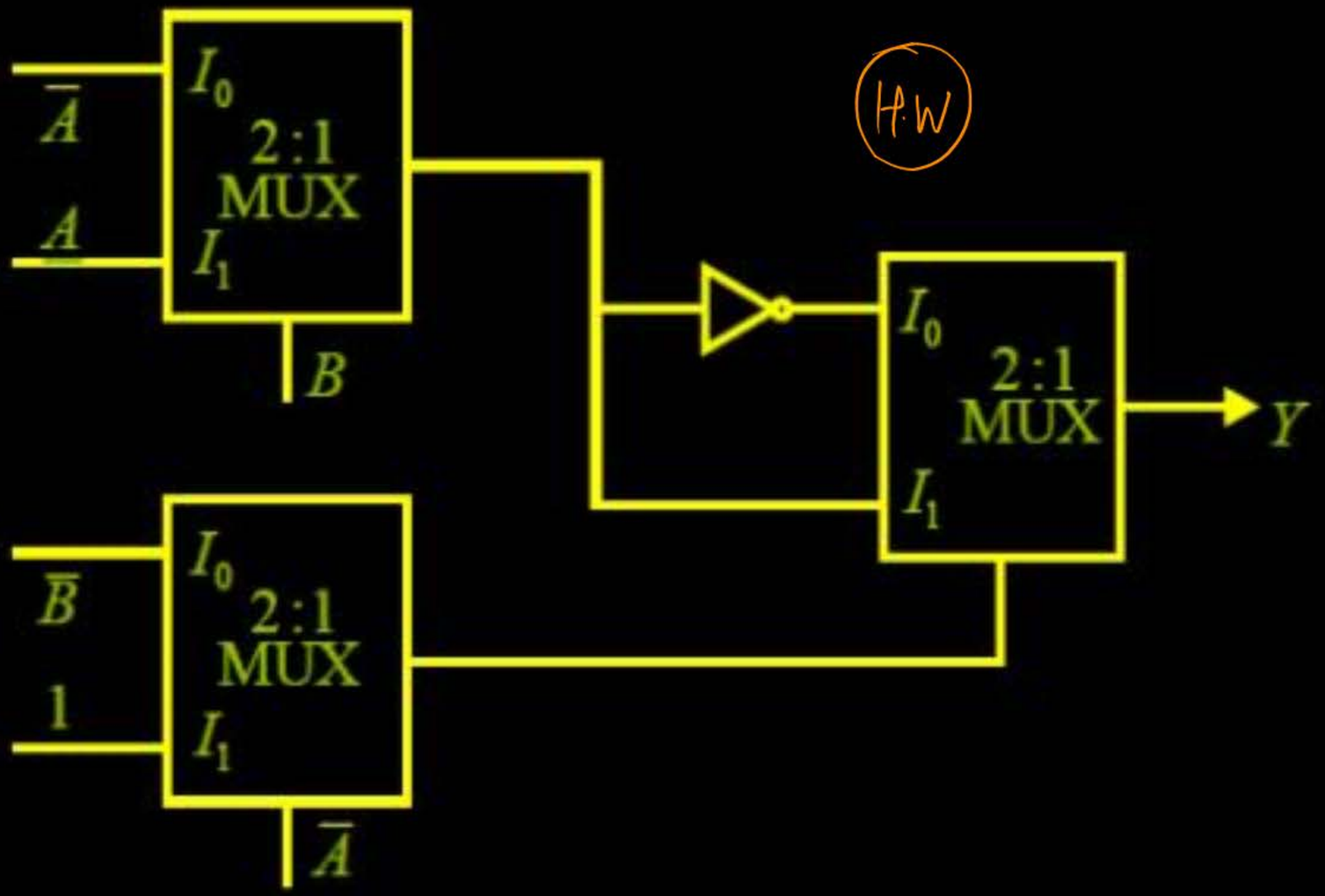
Output Y is

(a) 2 input NOR gate

(b) 2 input AND gate

(c) 2 input OR gate

(d) 2 input NAND gate



Question

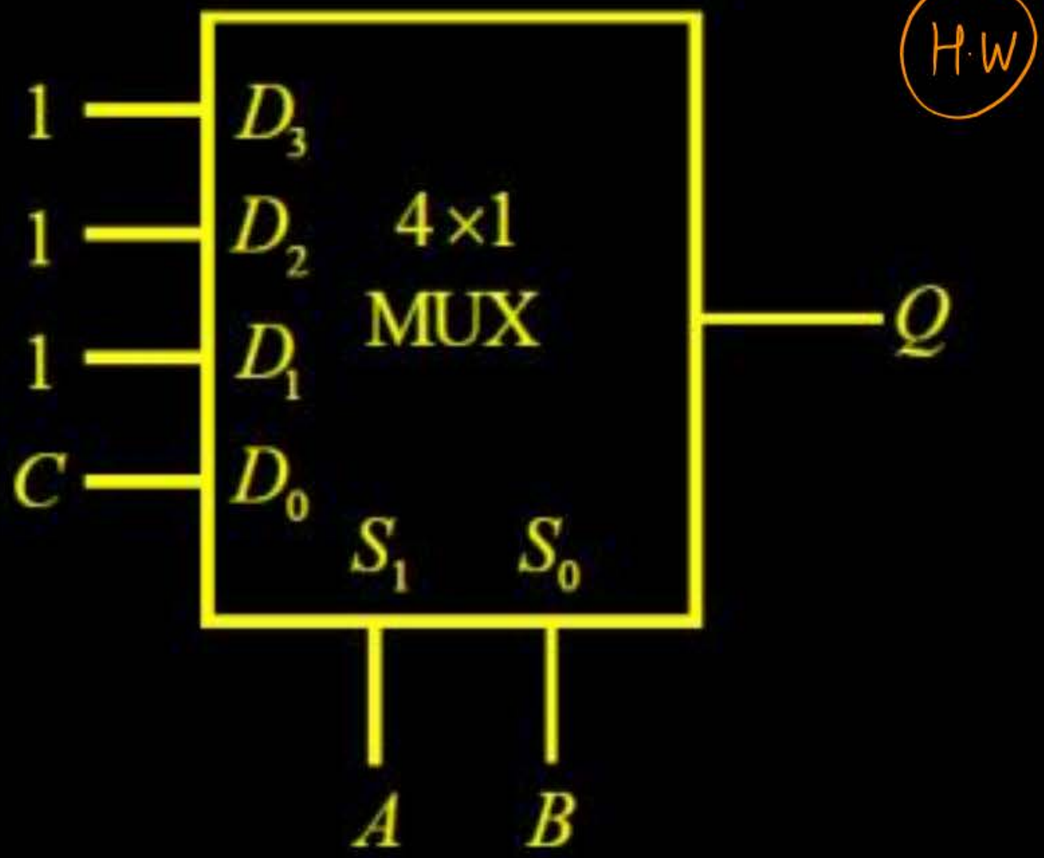
The combinational circuit shown in figure employs a 4 to 1 multiplexer. The output Q of the circuit is

(a) $\overline{A}\overline{B}C$

(b) $A + B + C$

(c) $A \oplus B \oplus C$

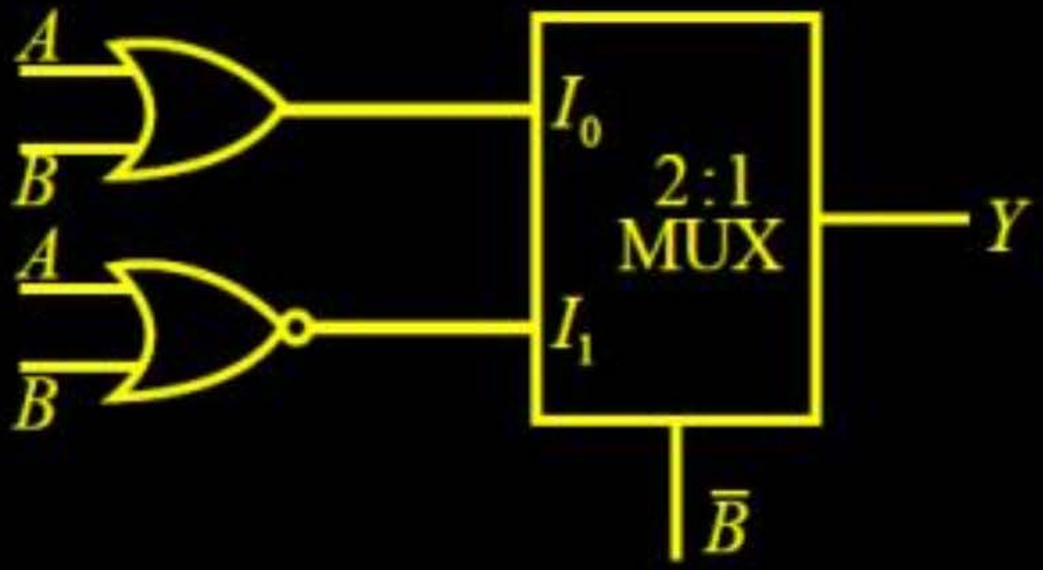
(d) $\overline{A}\overline{B}\overline{C}$



H·W

Question

Logical circuit is given as



H·W

Minimum no. of 2-input NAND gate require to implement Y is _____.

Question

For the identify $AB+\bar{A}C+BC=AB+\bar{A}C$, the dual form is

(a) $(A+B)(\bar{A}+C)(B+C)=(A+B)(\bar{A}+C)$

(b) $(\bar{A}+\bar{B})(A+\bar{C})(\bar{B}+\bar{C})=(\bar{A}+\bar{B})(A+\bar{C})$

(c) $(A+B)(\bar{A}+C)(B+C)=(\bar{A}+\bar{B})(A+\bar{C})$

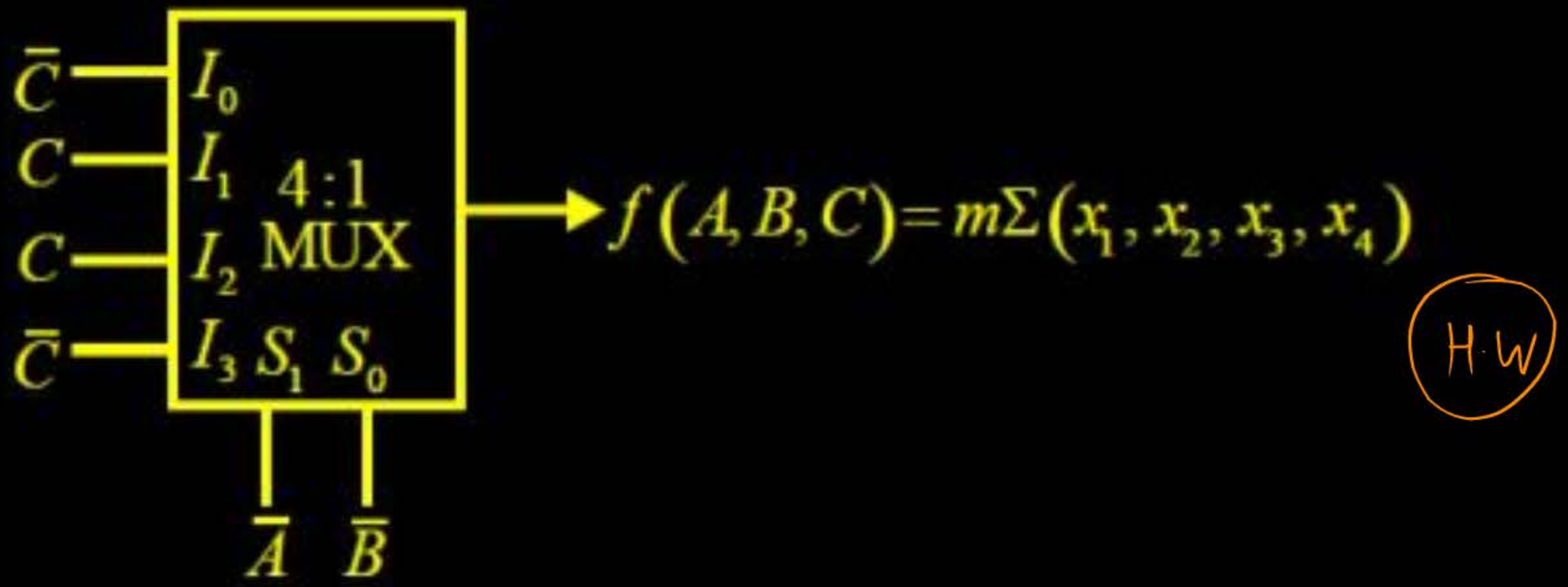
(d) $\bar{A}\bar{B}+A\bar{C}+\bar{B}\bar{C}=\bar{A}\bar{B}+A\bar{C}$

$(A+B)(\bar{A}+C)(B+C)$ → redundant

$=(A+B)(\bar{A}+C)$

Question

A 4:1 MUX is gives as:



H·W

Then $(x_1 + x_2 + x_3 + x_4)$ is_____.

Question

A digital circuit is as shown below:

When $f_1(A, B) = \Sigma(0, 1)$ $f_2(A, B) = \Sigma 1 = f_3(A, B) = \Sigma(0,2,3)$,

$f_4(A, B) = \Sigma(1,2,3)$ then $f(A, B, C)$ will be

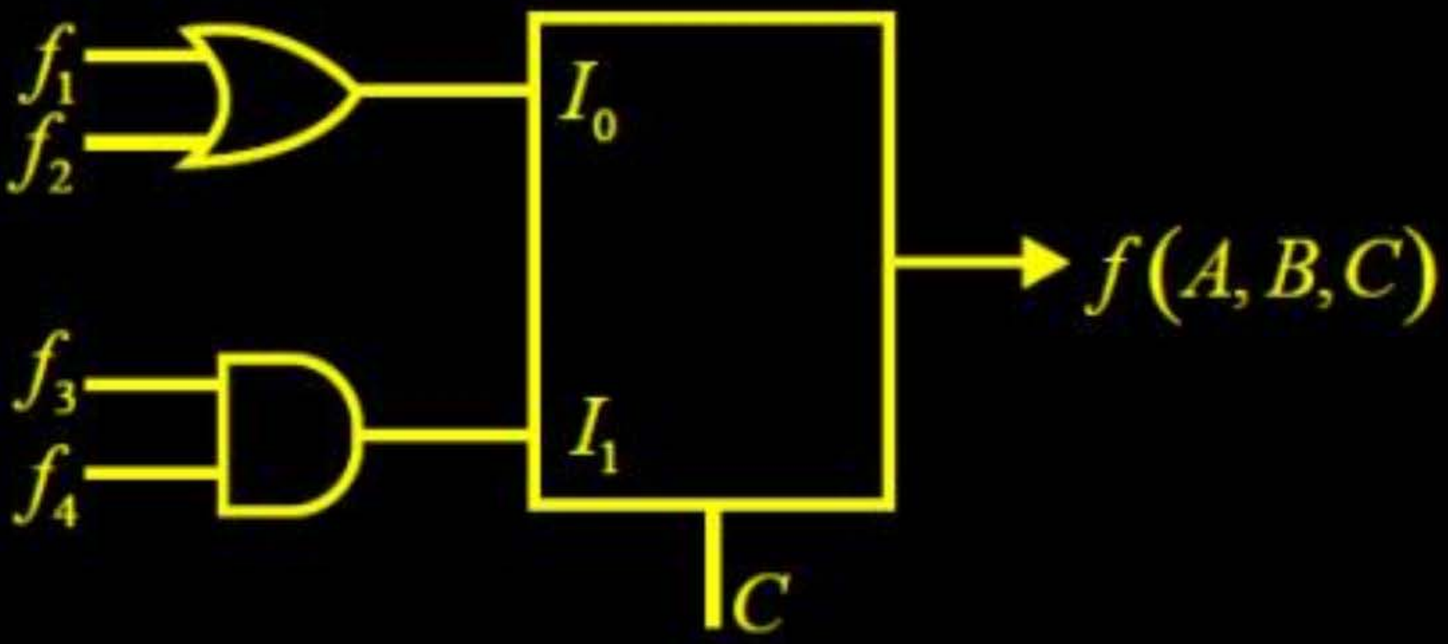
H.W

(a) $A \oplus C$

(b) $A \odot C$

(c) $A \oplus B \oplus C$

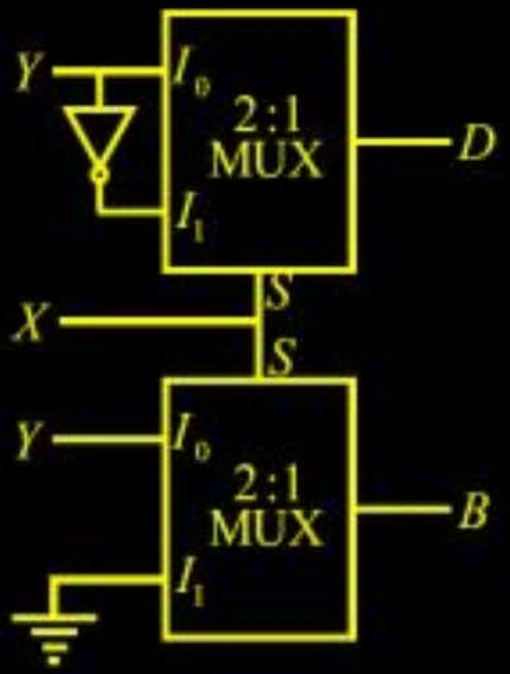
(d) $A \odot B \oplus C$



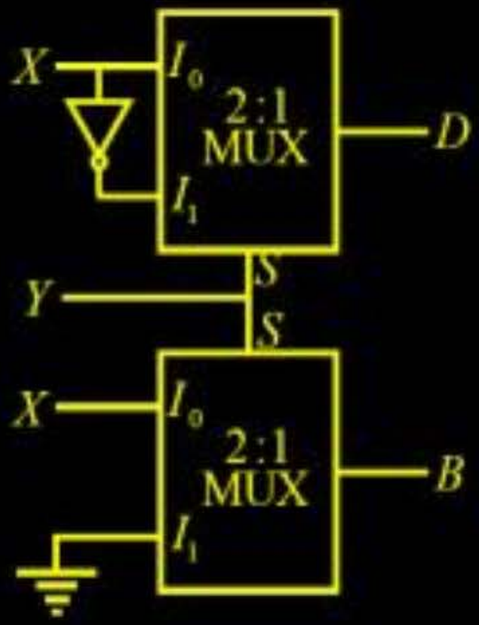
Question

If X and Y are inputs and the Difference (D = X – Y) and the Borrow (B) are the outputs, which one of the following diagrams implements a half-subtractor?

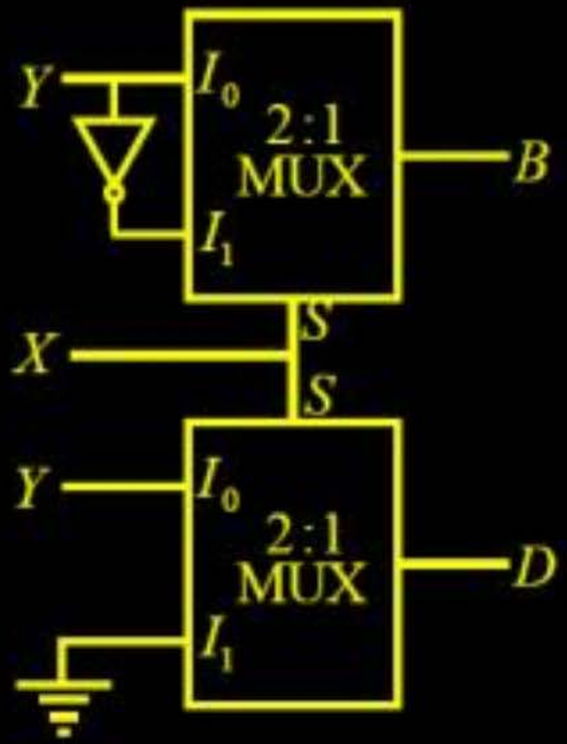
(a)



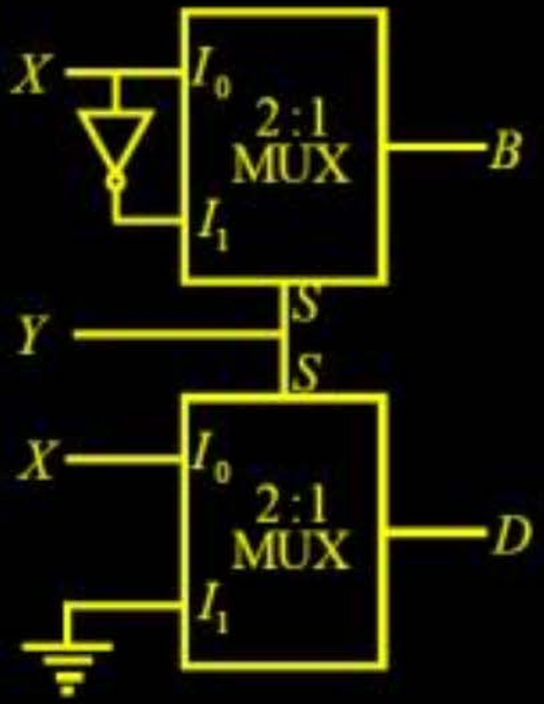
(b)



(c)

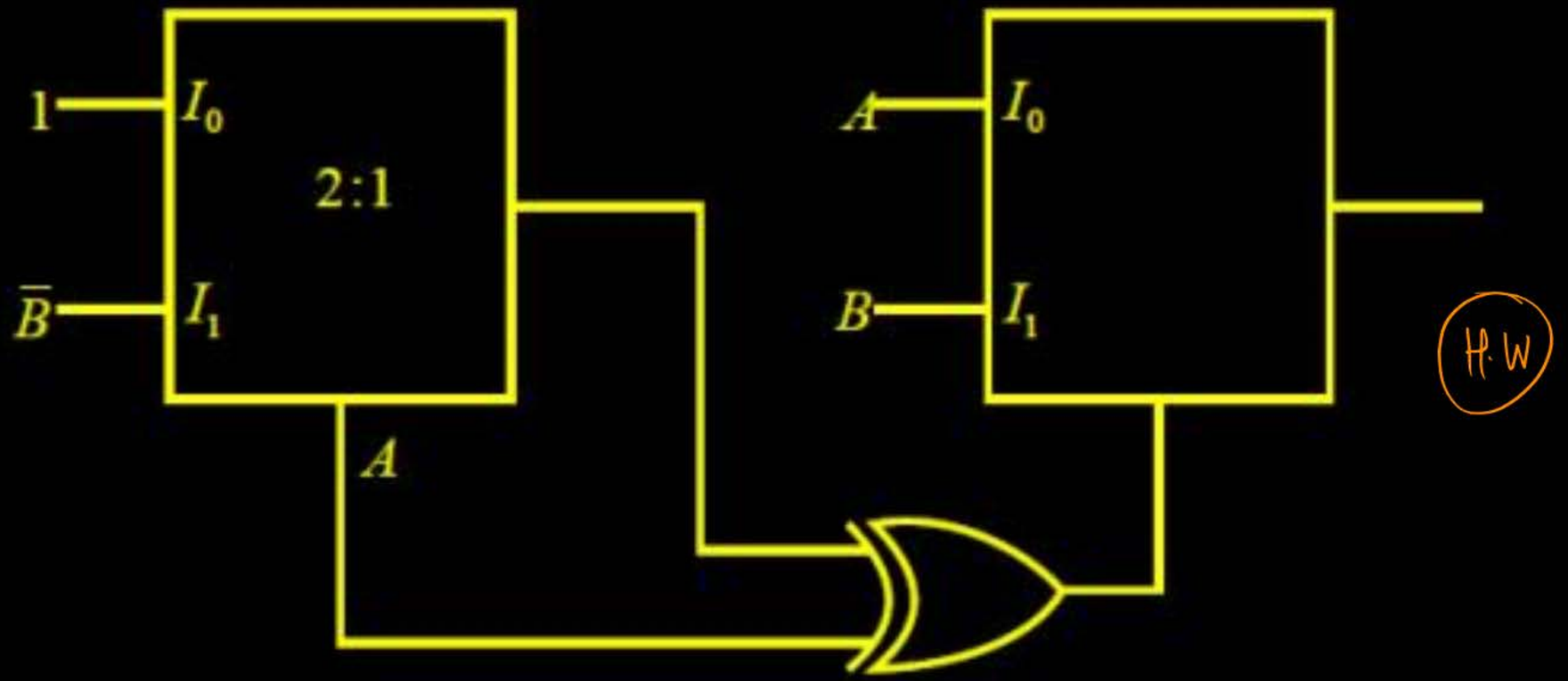


(d)



Question

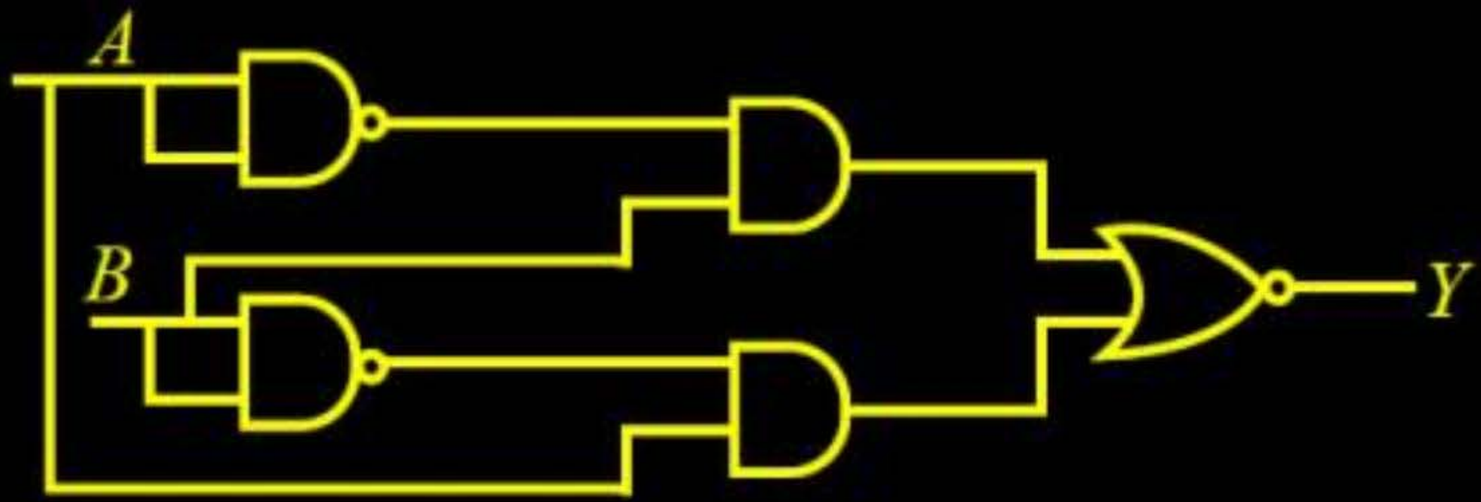
A logical circuit is as given below:



Minimum no. of two input NAND gate require to implement output Y is _____.

Question

A circuit is as given below:



H.W.

If all the gates are replaced by its dual gates in above circuit then output Y will changes to Y_1 and to implement Y_1 minimum no. of two input NAND gate required is _____.

Question

A 2-bit binary multiplier can be Implemented using

(a) 2 input AND gates only

(b) Six 2-input AND gates and two XOR gates.

(c) Two 2-input NORs and XNOR gate.

H·W

(d) XOR gates and shift registers



2 Minute Summary

Question Discussion

Thank you

GW

Soldiers !

